

C Programming Questions and Answers

1. Difference between while and do-while loop

while loop checks the condition before execution while do-while checks after execution. while loop may execute zero times but do-while executes at least once.

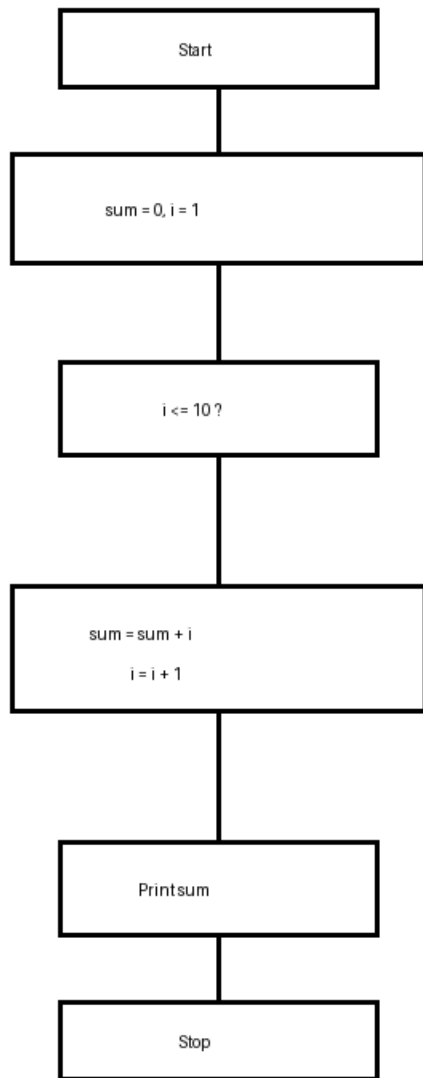
Example:

```
while(condition){ statements; }  
do{ statements; }while(condition);
```

2. Algorithm for summation of first 10 positive numbers

1. Start
2. Initialize sum = 0 and i = 1
3. Repeat while i <= 10
4. sum = sum + i
5. i = i + 1
6. Print sum
7. Stop

Flowchart:



3. Different Types of Operators in C

1. Arithmetic Operators: +, -, *, /, %
2. Relational Operators: ==, !=, >, <, >=, <=
3. Logical Operators: &&, ||, !
4. Assignment Operators: =, +=, -=, *=, /=
5. Increment/Decrement Operators: ++, --
6. Bitwise Operators: &, |, ^, ~, <<, >>
7. Conditional Operator: ?:

4. C Program to Print First 10 Even Numbers Using for Loop

```
#include <stdio.h>
int main()
```

```

{
    int i;
    for(i = 2; i <= 20; i += 2)
    {
        printf("%d\n", i);
    }
    return 0;
}

```

5. C Program to Print First 10 Odd Numbers Using while Loop

```

#include <stdio.h>
int main()
{
    int i = 1, count = 0;
    while(count < 10)
    {
        printf("%d\n", i);
        i += 2;
        count++;
    }
    return 0;
}

```

6. C Program to Check Whether a Number is Prime or Not

```

#include <stdio.h>
int main()
{
    int n, i, flag = 0;
    scanf("%d", &n);

    if(n <= 1)
        printf("Not Prime");
    else
    {
        for(i = 2; i <= n/2; i++)
        {
            if(n % i == 0)
            {
                flag = 1;
                break;
            }
        }

        if(flag == 0)
            printf("Prime");
        else
            printf("Not Prime");
    }

    return 0;
}

```

7. C Program to Check Whether an Alphabet is Vowel or Not Using switch-case

```

#include <stdio.h>

```

```
int main()
{
    char ch;
    scanf("%c", &ch);

    switch(ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            printf("Vowel");
            break;

        default:
            printf("Not Vowel");
    }

    return 0;
}
```